

Mar. 06
Part II

1. Heavenly Skin Cosmetics is testing two creams (Cream-A and Cream-B) to see which is better for reducing the number of freckles on skin.. A total of 8 people are tested; 4 are randomly chosen to get Cream A and the other 4 to get Cream B. One week after the cream is applied, the number of freckles on the left hand are counted for all 8 patients with the following results:

The Patients Receiving Cream A: 7 freckles, 8 freckles, 9 freckles, 10 freckles

The patients receiving Cream B: 11 freckles, 12 freckles, 13 freckles, 99 freckles

We will initially use a two-sample t-test to test the following hypothesis

$$H_0: \mu_A = \mu_B$$

$$H_A: \mu_A < \mu_B$$

- A. State the assumptions needed to do a two sample t-test *and assume for now that these assumptions are valid*

- B. Calculate the Test-Statistic

- C. Use the Test-Statistic to Evaluate the null hypothesis at $\alpha = 0.05$

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